

The Liquidity Channel of Fiscal Policy

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Motivation

- After the 2020 recession, public debt climbs to an unprecedented level.
- It is important to understand the effects of public debt on the economy.
- An essential aspect of public debt is private liquidity, so this paper quantifies the liquidity role of public debt.

Empirical Part

- Fiscal policy has a sizable impact on the return difference between public debt and less liquid assets.
 - 1% increase in US public debt reduces real estate return by 35 basis points relative to the government bond.

Quantitative Part

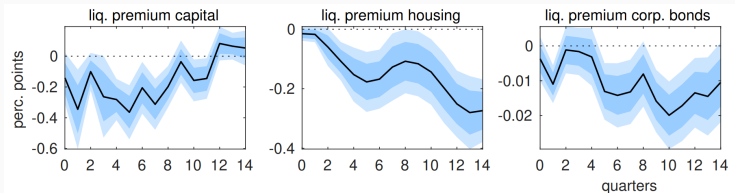
- Rationalize the empirical finding with a HANK model.
 - Asset classes vary in their degree of liquidity.
- Public debt reduces the liquidity premium.
 - Public debt increases the supply of liquid savings devices.
 - HH holds these additional liquid savings only when the liquidity premium falls.
- The movement in liquidity premium increases the response of output to the fiscal stimulus.

Evidence from Local Projections

- Fiscal expansions reduce liquidity premium, i.e., the return differences between public debt and less liquid assets.
- Changes in public debt are endogenous to other shocks.
- Look at exogenous government spending shocks identified Blanchard and Perotti(2002).
 - Government spending is predetermined within the quarter, so it is exogenous to other shocks in quarterly data.

$$x_{t+h} = \beta_0 + \beta_1 t + \beta_2 t^2 + \phi_h \log g_t + \Gamma(L)Z_{t-1} + u_{t+h}$$

- x_{t+h} is the liquidity premium.
 - Government bond is 20-Year Treasury Constant Maturity Rate.
- g_t is real government spending per capita.
- Under Blanchard and Perotti(2002), ϕ_h provides the impulse response to the government spending shock in t .



- The government expenditure shock is scaled so that debt responds by 1 percent.
- The fiscal expansion goes along with a significant fall in all liquidity premium measures.
- The return of capital is from Gomme et al. (2011). The return of housing is from Jorda et al. (2019). Corporate bonds are AAA-rated.

The model is composed of a HH sector, a firm sector, and a government sector.

Households

- Two types of agents: workers and entrepreneurs. The transition is stochastic.
- Only workers supply labor and face labor-income risks. Entrepreneurs earn pure rents.
- All HH self-insured by liquid (bonds) and illiquid (capital) assets. Trading illiquid assets is subject to random participation.

- HH's labor income, $w_t n_{it} h_{it}$, is composed of aggregate wage, w_t , hours worked, n_{it} , and labor productivity h_{it} . Entrepreneur's productivity (h_{it}) is 0.

Worker's productivity

$$h_{it} = \begin{cases} \exp(\rho_h \log h_{it-1} + \epsilon_{it}^h) & \text{w.p. } 1 - \zeta \\ 0 & \text{w.p. } \zeta \end{cases}$$

Entrepreneur's productivity

$$h_{it} = \begin{cases} 1 & \text{w.p. } \iota \\ 0 & \text{w.p. } 1 - \iota \end{cases}$$

HH preference

$$E \max_{\{c_{it}, n_{it}\}} \sum_{t=0}^{\infty} \beta^t u[c_{it} - L(h_{it}, n_{it})]$$

where $L(h_{it}, n_{it})$ is labor disutility.

The utility function is CRRA. $u(x_{it}) = \frac{1}{1-\xi} x_{it}^{1-\xi}$.

Goods consumption bundles varieties j of differentiated goods

$$c_{it} = \left(\int c_{ijt}^{\frac{\eta_t-1}{\eta_t}} dj \right)^{\frac{\eta_t}{\eta_t-1}}.$$

Labor disutility is $L(h_{it}, n_{it}) = h_{it} \frac{n_{it}^{1+\gamma}}{1+\gamma}$

Consumption and Portfolio choice

- Portfolio choice between liquid bonds and illiquid capital. Only λ fraction of HH can adjust their capital.
- There is a wedge (A_t) between government bond yield (R_t^b) and interest paid to HH (R_t).

$$R(b_{it}, R_t^b, A_t) = \begin{cases} R_t^b A_t & \text{if } b_{it} \geq 0 \\ R_t^b A_t + \bar{R} & \text{if } b_{it} < 0 \end{cases}$$

- The extra wedge creates a mass of HH with zero credit.

Budget constraint

$$\begin{aligned}c_{it} + b_{it+1} + q_t k_{it+1} &= (1 - \tau)(h_{it} w_t N_t + \mathbf{1}_{h_{it} \neq 0} \Pi_t^U + \mathbf{1}_{h_{it} = 0} \Pi_t^F) \\ &+ T_t(h_{it}) + b_{it} \frac{R(b_{it}, R_t^b, A_t)}{\pi_t} + (q_r + r_t) k_{it}, \\ b_{it+1} &\geq \bar{B}, \quad k_{it+1} \geq 0.\end{aligned}$$

b_{it} is real bond, k_{it} is illiquid assets, Π_t^U is union profits, Π_t^F is firm profits, $T_t(h_{it})$ is non-distortionary transfers, $R(b_{it}, R_t^b, A_t)$ is nominal interest rate schedule on bond.

Firm sector

Firm sector consists of four subsectors.

1. Labor sector: Unions and Labor packers.
2. Intermediate goods producers
3. Final goods producers
4. Capital goods producers

Unions

- Unions indexed by j buy labor services from HH and sell to labor packers.
- Unions maximize profits by setting W_{jt} given W_t (unions buy labor from HH) and W_t^F (labor packers sell labor services to intermediate goods producers).

$$E \sum_{t=0}^{\infty} \beta^t \lambda_w^t \frac{W_t^F}{P_t} N_t \left\{ \left(\frac{W_{jt} \bar{\pi}_W^t}{W_t^F} - \frac{W_t}{W_t^F} \right) \left(\frac{W_{jt} \bar{\pi}_W^t}{W_t^F} \right)^{-\zeta_t} \right\}$$

- Obtains wage Phillips curve

$$\log\left(\frac{\pi_t^W}{\bar{\pi}^W}\right) = \beta E_t \log\left(\frac{\pi_{t+1}^W}{\bar{\pi}^W}\right) + \kappa_w \left(\frac{w_t}{w_t^F} - \frac{\zeta_t - 1}{\zeta_t} \right)$$

Labor packers

- Labor packers buy differentiated labors from unions and produce final labor services.

$$N_t = \left(\int \hat{n}_{jt}^{\frac{\zeta_t-1}{\zeta_t}} dj \right)^{\frac{\zeta_t}{\zeta_t-1}}$$

- They sell final labor services to intermediate goods producers at wage W_t^F .

Final Goods Producers

- Final goods producers face $y_{jt} = (\frac{p_{jt}}{P_t})^{-\eta_t} Y_t$ for each goods j .
- They choose p_{jt} to maximize

$$E_0 \sum_{t=0}^{\infty} \beta^t \lambda_Y^t (1 - \tau) Y_t \left\{ \left(\frac{p_{jt} \bar{\pi}_Y^t}{P_t} - \frac{MC_t}{P_t} \right) \left(\frac{p_{jt} \bar{\pi}^t}{P_t} \right)^{-\eta_t} \right\}$$

- Obtain Phillips curve

$$\log\left(\frac{\pi_t}{\bar{\pi}}\right) = \beta E_t \log\left(\frac{\pi_{t+1}}{\bar{\pi}}\right) + \kappa_Y \left(mc_t - \frac{\eta_t - 1}{\eta_t} \right)$$

Intermediate goods producers

- Intermediate goods are produced with $Z_t N_t^\alpha (u_t K_t)^{1-\alpha}$.
- They choose N_t, K_t, u_t to maximize

$$m c_t Z_t N_t^\alpha (u_t K_t)^{1-\alpha} - w_t^F N_t - [r_t + q_t \delta(u_t)] K_t$$

where $\delta(u_t) = \delta_0 + \delta_1(u_t - 1) + \delta_2/2(u_t - 1)^2$.

Capital Goods Producers

- Capital goods producers decide I_t given capital price q_t

$$E_0 \sum_{t=0}^{\infty} \beta^t I_t \left\{ \Psi_t q_t \left[1 - \frac{\phi}{2} \left(\log \frac{I_t}{I_{t-1}} \right)^2 \right] \right\},$$

where Ψ_t governs the marginal efficiency of investment.

- The law of motion for aggregate capital

$$K_t - (1 - \delta(u_t))K_{t-1} = \Psi_t \left[1 - \frac{\phi}{2} \left(\log \frac{I_t}{I_{t-1}} \right)^2 \right] I_t.$$

Government sector

The government expenditure rule

$$\frac{G_t}{\bar{G}} = \left(\frac{G_{t-1}}{\bar{G}} \right)^{\rho_G} \left(\frac{Y_t}{Y_{t-1}} \right)^{(1-\rho_G)\gamma_Y} \left(\frac{B_t}{\bar{B}} \right)^{(1-\rho_G)\gamma_B} D_t$$

- D_t is a government spending shock.
- Tax revenues are $T_t = \tau(w_t N_t + \Pi_t^U + \Pi_t^F) - T_t$.
- Government bonds are determined residually.

$$B_{t+1} = G_t - T_t + \frac{R_t^b}{\pi} B_t$$

- Monetary policy rule is

$$\frac{R_{t+1}^b}{\bar{R}} = \left(\frac{R_t^b}{\bar{R}^b} \right)^{\rho_R} \left(\frac{\pi_t}{\bar{\pi}} \right)^{(1-\rho_R)\theta_\pi} \left(\frac{Y_t}{Y_{t-1}} \right)^{(1-\rho_R)\theta_y} \epsilon_t^R$$

Market Clearing

The bond market clears when the supply of government bonds is equal to the total net amount of HH's bonds.

$$B_{t+1} = E_t[\lambda b_{at} + (1 - \lambda)b_{nt}]$$

The private liquidity is the sum over all private debt, defined as $IOU_t = \int_{\bar{B}}^0 b d\Theta_t$.

The capital market clearing

$$K_{t+1} = E_t[\lambda k_t^* + (1 - \lambda)k]$$

Calibration and Estimation

- First, calibrate all parameters that affect SS of the model.
- Second, estimate all parameters that matter for dynamics by Bayesian full-information methods.

Table 2: Calibration Targets

Targets	Model	Data	Source	Parameter
Mean illiquid assets (K/Y)	11.48	11.48	NIPA	Discount factor
Public liquidity (B/Y)	2.36	2.36	FRED	Port. adj. probability
Private liquidity (IOUs/Y)	0.56	0.56	FOF	Borrowing penalty
Top 10% wealth share	0.68	0.68	WID	Fraction of entrepreneurs

table

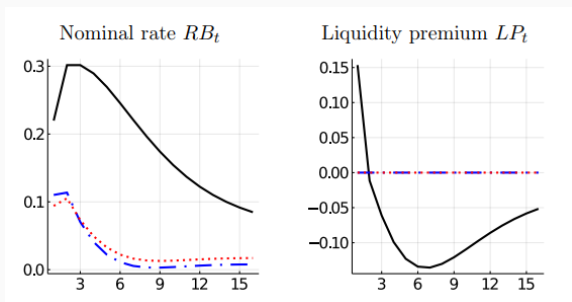
Estimation

- Estimate remaining parameters that matter only for the dynamics of the model by Bayesian full-information methods.
- Data: US data from 1947Q1 to 2019Q4.
- Growth rate of GDP per capita, consumption, investment, wages, GDP deflator, hours worked, federal funds rate, liquidity premium.
- Parameter estimates are in line with RA literature.

table2

Government spending shocks

- Transitory government spending shocks that increase 1 percent debt at the peak.
- Liquidity premium declines after shocks.
- Local projections show the decline is between 2 and 35 basis points.



Why does liquidity premium decline?

- Increased public spending crowds out capital.
 - Resources are absorbed by the government, but consumption falls less. → Investment falls.
 - Further savings become available.
- Private liquidity falls only by 0.4%.
- Total liquidity increases because the increase in government bonds is larger than the decline in private liquidity.
- The bond rate rises to absorb the extra liquidity.
- The bond rate increases more than capital return, so the liquidity premium falls.

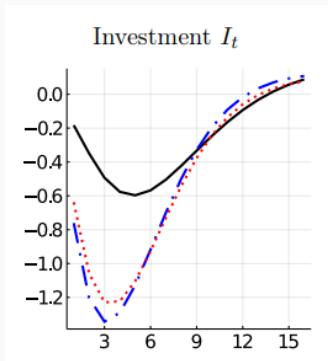
How much does the liquidity injection amplify the fiscal multiplier?

- Baseline model vs. Liquid-capital model vs. RANK
 - In the liquid-capital model, HH can always adjust capital.
- Fiscal Multiplier: 1.28 vs. 0.77 vs. 1.11

Why fiscal multiplier is larger?

- The increase in liquidity improves the self-insurance of households, boosting consumption.
- Investment drops larger in the liquidity-capital model & RANK.
- In the liquid-capital model, there is no movement in the liquidity premium.

- Baseline model: With portfolio choice, the liquidity premium falls. There is less capital crowding out.
- Liquidity-capital model & RANK: The liquidity premium is high after spending shocks. There is more crowding out.



Comments

- In the empirical part, they focus on government spending shocks that increase public debt.
- The effect of government spending and public debt is not identified.
- Participation in the capital market is exogenous.

A *sequential equilibrium with recursive planning* in our model is a sequence of policy functions $\{x_{a,t}^*, x_{n,t}^*, b_{a,t}^*, b_{n,t}^*, k_t^*\}$, of value functions $\{V_t^a, V_t^n\}$, of prices $\{w_t, w_t^F, \Pi_t^F, \Pi_t^U, q_t, r_t, R_t^b, \pi_t, \pi_t^W\}$, of stochastic states A_t, Ψ_t, Z_t and shocks $\epsilon_t^R, \epsilon_t^G, \epsilon_t^A, \epsilon_t^Z, \epsilon_t^\Psi, \epsilon_t^{\mu W}, \epsilon_t^{\mu Y}$, aggregate capital and labor supplies $\{K_t, N_t\}$, distributions Θ_t over individual asset holdings and productivity, and expectations Γ for the distribution of future prices, such that

1. Given the functional $\mathbb{W}_{t+1}$ for the continuation value and period- t prices, policy functions $\{x_{a,t}^*, x_{n,t}^*, b_{a,t}^*, b_{n,t}^*, k_t^*\}$ solve the households' planning problem, and given the policy functions $\{x_{a,t}^*, x_{n,t}^*, b_{a,t}^*, b_{n,t}^*, k_t^*\}$ and prices, the value functions $\{V_t^a, V_t^n\}$ are a solution to the Bellman equation (7).
2. Distributions of wealth and income evolve according to households' policy functions.
3. The labor, the final goods, the bond, the capital, and the intermediate goods markets clear in every period, interest rates on bonds are set according to the central bank's Taylor rule, fiscal policy is set according to the fiscal rule, and stochastic processes evolve according to their laws of motion.
4. Expectations are model consistent.

Parameter	Value	Description	Target
Households			
β	0.983	Discount factor	see Table 2
ξ	4.00	Relative risk aversion	Kaplan et al. (2018)
γ	2.00	Inverse of Frisch elasticity	Chetty et al. (2011)
λ	6.40%	Portfolio adj. prob.	see Table 2
ρ_h	0.98	Persistence labor income	Storesletten et al. (2004)
σ_h	0.12	STD labor income	Storesletten et al. (2004)
ζ	0.05%	Trans.prob. from W. to E.	see Table 2
ι	6.25%	Trans.prob. from E. to W.	Güvener et al. (2014)
$\bar{R}$	1.00%	Borrowing penalty	see Table 2
Firms			
α	0.68	Share of labor	62% labor income
δ_0	1.75%	Depreciation rate	7.0% p.a.
$\bar{\eta}$	11	Elasticity of substitution	Price markup 10%
$\bar{\zeta}$	11	Elasticity of substitution	Wage markup 10%
Government			
τ	0.28	Tax rate level	$G/Y = 20\%$
$\bar{R}^b$	1.00	Nominal rate	see text
$\bar{\pi}$	1.00	Inflation	see text

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Parameter	Distribution	Prior		Posterior			
		Mean	Std. Dev.	Mean	Std. Dev.	5 %	95 %
Frictions							
δ_s	Gamma	5.00	2.00	0.499	0.086	0.363	0.645
ϕ	Gamma	4.00	2.00	0.117	0.017	0.090	0.146
κ	Gamma	0.10	0.01	0.111	0.010	0.096	0.128
κ_w	Gamma	0.10	0.01	0.099	0.011	0.082	0.117
Monetary policy rule							
ρ_R	Beta	0.50	0.20	0.820	0.012	0.800	0.839
σ_R	Inv.-Gamma	0.10	2.00	0.230	0.011	0.213	0.250
θ_π	Normal	1.70	0.30	1.704	0.086	1.567	1.846
θ_Y	Normal	0.13	0.05	0.360	0.041	0.293	0.428
Spending rule							
γ_B	Normal	0.00	1.00	-0.335	0.094	-0.493	-0.186
γ_Y	Normal	0.00	1.00	-9.925	0.637	-11.00	-8.902
ρ_G	Beta	0.50	0.20	0.727	0.024	0.686	0.766
γ_ϵ	Beta	0.50	0.20	0.385	0.043	0.312	0.454
σ_G	Inv.-Gamma	0.10	2.00	4.689	0.285	4.245	5.178
Structural shocks							
ρ_A	Beta	0.50	0.20	0.976	0.008	0.961	0.989
σ_A	Inv.-Gamma	0.10	2.00	0.177	0.018	0.148	0.207
ρ_Z	Beta	0.50	0.20	0.972	0.006	0.963	0.981
σ_Z	Inv.-Gamma	0.10	2.00	0.803	0.035	0.749	0.862
ρ_Ψ	Beta	0.50	0.20	0.936	0.008	0.922	0.950
σ_Ψ	Inv.-Gamma	0.10	2.00	1.868	0.112	1.692	2.058
ρ_μ	Beta	0.50	0.20	0.854	0.018	0.824	0.882
σ_μ	Inv.-Gamma	0.10	2.00	1.938	0.138	1.723	2.174
$\rho_{\mu w}$	Beta	0.50	0.20	0.786	0.026	0.742	0.828
$\sigma_{\mu w}$	Inv.-Gamma	0.10	2.00	7.175	0.757	6.036	8.516
Measurement error							
σ_{LP}^{me}	Inv.-Gamma	0.05	0.01	1.885	0.091	1.741	2.040

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